

BELE-001: Basic of Electrical & Electronics Engineering (BEEE)

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UNIT I (25%)

DC Circuits, KCL/KVL, Network Theorems, AC Sinusoidal Waveforms, Phasors & Resonance.

UNIT II (25%)Magnetic Circuits, B-H Loop, 1- Φ Transformers, OC/SC Tests, DC Machines & Induction Motors.**UNIT III (25%)**

P-N Junctions, Bridge Rectifiers, Clippers/Clampers, Zener Regulators & BJT CE Characteristics.

UNIT IV (25%)

Number Systems, Binary Codes, Logic Gates, De Morgan's & 4-Variable K-Map Minimization.

UNIT I: DC CIRCUITS, NETWORK THEOREMS & SINGLE-PHASE AC CIRCUITS

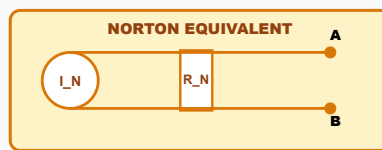
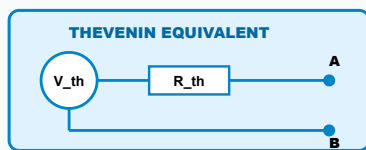
Circuit Elements, KCL, KVL, Thevenin, Norton, Max Power Transfer, AC Phasors, Power Triangle & RLC Resonance

1. Circuit Elements, Kirchhoff's Laws & Network Theorems

Circuit Elements: Resistor (linear energy dissipator $V = IR$), Inductor (stores magnetic field energy $v(t) = L (di/dt)$, $E = 0.5 L I^2$), Capacitor (stores electrostatic field energy $i(t) = C (dv/dt)$, $E = 0.5 C V^2$).

Kirchhoff's Laws:

- Kirchhoff's Current Law (KCL):** Based on conservation of charge. Algebraic sum of all currents entering and exiting a node is zero: $\sum I = 0$.
- Kirchhoff's Voltage Law (KVL):** Based on conservation of energy. Algebraic sum of all potential differences around any closed circuit loop is zero: $\sum V = 0$.

**FIGURE 1.1: THEVENIN AND NORTON DUAL NETWORK MODELS**

Network Theorem	Core Statement & Formal Mechanism	Terminal Boundary Equations
Superposition Theorem	In any linear bilateral circuit containing multiple independent sources, the response (voltage/current) across any element equals the algebraic sum of responses caused by each source acting alone, with all other independent voltage sources short-circuited and current sources open-circuited.	$I_{total} = \sum_{k=1}^n I_k$ Condition: Linear circuits only (does not apply directly to non-linear power $P \neq \sum P_k$).
Thevenin's Theorem	Any two-terminal linear network containing independent sources and resistances can be replaced by an equivalent circuit consisting of an ideal voltage source V_{th} in series with a resistance R_{th} .	$V_{th} = V_{oc}$ (Open-circuit voltage at terminals A-B) $R_{th} = R_{in}$ (Input resistance with sources de-energized).
Norton's Theorem	Dual of Thevenin: Any two-terminal active linear network can be replaced by an ideal current source I_N connected in parallel with a resistance $R_N = R_{th}$.	$I_N = I_{sc}$ (Short-circuit current across terminals A-B) $I_N = V_{th} / R_{th}$.
Maximum Power Transfer	A DC source delivers maximum power to a variable load resistance R_L when the load resistance equals the Thevenin internal resistance of the network as viewed from load terminals.	Condition: $R_L = R_{th}$ $P_{max} = V_{th}^2 / (4 R_{th})$ Efficiency at maximum power transfer = 50%.

2. Single-Phase AC Circuits, Power Triangle & RLC Resonance

Sinusoidal Waveform: $v(t) = V_m \sin(\omega t + \phi)$. Angular frequency $\omega = 2\pi f$. RMS Value: $V_{rms} = V_m / \sqrt{2} = 0.707 V_m$. Average Value: $V_{avg} = \frac{2}{\pi} V_m = 0.637 V_m$. Form Factor $K_f = V_{rms}/V_{avg} = 1.11$. Peak Factor $K_p = V_m / V_{rms} = 1.414$.

Complex Power & Power Triangle:

- **Active / Real Power (P):** Work-producing power dissipated in resistance. $P = V I \cos\phi$ (Watts or kW).
- **Reactive Power (Q):** Energy oscillating between source and reactive elements (L, C). $Q = V I \sin\phi$ (VAR or kVAR).
- **Apparent Power (S):** Total complex power supplied by source. $S = V I = \sqrt{P^2 + Q^2}$ (VA or kVA).
- **Power Factor (pf):** Ratio of active power to apparent power: $pf = \cos\phi = P / S = R / |Z|$.

Series RLC Resonance:

Condition: $X_L = X_C \Rightarrow \omega_0 L = 1 / (\omega_0 C) \Rightarrow f_0 = 1 / (2\pi\sqrt{LC})$.

- At resonance, net reactance is zero: $Z = R + j(X_L - X_C) = R$ (Minimum Impedance $\Rightarrow$ Maximum Current $I_0 = V / R$).
- Power factor is strictly UNITY ($\cos\phi = 1.0$).
- **Quality Factor (Q):** Voltage magnification factor across reactive components: $Q = (\omega_0 L) / R = 1 / (\omega_0 C R) = \frac{1}{R} \sqrt{L/C}$.
- **Bandwidth (BW):** $BW = f_2 - f_1 = f_0 / Q = R / (2\pi L)$.

UNIT II: TRANSFORMERS & ELECTRICAL ROTATING MACHINES

Magnetic Circuits, B-H Loop, 1-Φ Transformer, OC/SC Tests, DC Machines & 3-Phase Induction Motors

1. Single-Phase Transformer: Principles, Equivalent Circuit & Testing

A static electrical machine that transfers electrical energy between circuits through mutual electromagnetic induction without changing frequency. EMF Equation: $E_1 = 4.44 f N_1 \Phi_m$; $E_2 = 4.44 f N_2 \Phi_m$.

Transformation Ratio: $K = E_2 / E_1 = N_2 / N_1 = I_1 / I_2$.

Loss Classification	Physical Origin & Governing Formula	Mitigation Technique in Manufacturing
Hysteresis Loss (P_h)	Energy dissipated as heat during cyclic reversal of magnetic domains in core: $P_h = k_h \cdot f \cdot B_m^{1.6} \cdot V$.	Use high-grade Silicon Steel (CRGO: Cold Rolled Grain Oriented) with narrow hysteresis loop.
Eddy Current Loss (P_e)	Circulating eddy currents induced in conductive core body by alternating flux: $P_e = k_e \cdot f^2 \cdot B_m^2 \cdot t^2 \cdot V$.	Laminate the core into thin varnished sheets (thickness $t = 0.35$ mm to 0.5 mm) to break current paths.
Copper Loss (P_{cu})	Ohmic $I^2 R$ heating loss in primary and secondary copper windings: $P_{cu} = I_1^2 R_1 + I_2^2 R_2 = I_1^2 R_{01}$.	Optimize conductor cross-sectional area with high-conductivity electrolytic copper.

Condition for Maximum Transformer Efficiency:

$$\text{Efficiency } \eta = \frac{V_2 I_2 \cos\phi}{V_2 I_2 \cos\phi + P_i + I_2^2 R_{02}}$$

Differentiating η with respect to load current I_2 and setting to zero yields: $P_i = I_2^2 R_{02} = P_{cu}$.

Maximum efficiency occurs when Variable Copper Loss strictly equals Constant Iron Loss!

Load current for max efficiency: $I_{2,max} = I_2 f_1 \cdot \sqrt{P_i / P_{cu} f_1}$.

2. DC Machines & Polyphase Induction Motors

DC Generator & Motor Duality:

- EMF Equation:** $E = \frac{\Phi Z N P}{60 A}$, where $A = 2$ for Wave winding and $A = P$ for Lap winding.
- Torque Equation:** $T = \frac{1}{2\pi} \cdot (\Phi Z I_a P / A) = 0.159 \Phi Z I_a (P/A) \text{ N}\cdot\text{m}$.
- Back EMF:** In a motor, $E_b = V - I_a R_a$ acts as a natural governor regulating armature current: $I_a = (V - E_b) / R_a$.
- Speed Control of DC Shunt Motor:**
 - Armature Resistance Control:* Adding external resistance in armature circuit drops voltage across armature, controlling speed **below rated speed**.
 - Field Flux Control:* Weakening field flux via rheostat reduces Φ , increasing speed **above rated speed**.

Three-Phase Induction Motor:

- Synchronous Speed:** $N_s = (120 f) / P \text{ rpm}$.
- Rotor Slip (s):** Normalized difference between synchronous and actual rotor speed: $s = (N_s - N) / N_s$.
- Rotor Frequency:** $f_r = s \cdot f$.
- Torque-Slip Relationship:** At low slip (normal operating zone), torque is directly proportional to slip: $T \propto s$ (linear stable region). At high slip (near standstill), torque is inversely proportional to slip: $T \propto 1/s$ (unstable region). Maximum torque occurs at slip $s_{max} = R_2 / X_2$.

UNIT III: SEMICONDUCTOR DEVICES, RECTIFIERS & BIPOLAR JUNCTION TRANSISTORS

Band Theory, P-N Junction V-I, Rectifier Analysis, Zener Regulator & BJT CE Characteristics

1. Semiconductor Band Theory, P-N Diode & Rectifier Analysis

Energy Band Theory: Conductors have overlapping valence and conduction bands; Insulators have large forbidden energy gaps ($E_g > 5$ eV); Semiconductors have small forbidden gaps ($E_g \approx 1.1$ eV for Si, 0.67 eV for Ge).

P-N Junction Diode: Forward bias narrows depletion layer allowing exponential diffusion current: $I = I_o (e^{V / \eta V_T} - 1)$. Reverse bias widens depletion region, leaving tiny reverse saturation current I_o .

Rectifier Parameter	Half-Wave Rectifier (HWR)	Center-Tapped Full-Wave	Full-Wave Bridge Rectifier
Number of Diodes	1	2	4
DC Voltage (Vdc)	$V_m / \pi \approx 0.318 V_m$	$2 V_m / \pi \approx 0.636 V_m$	$2 V_m / \pi \approx 0.636 V_m$
RMS Voltage (Vrms)	$V_m / 2 = 0.5 V_m$	$V_m / \sqrt{2} = 0.707 V_m$	$V_m / \sqrt{2} = 0.707 V_m$
Rectification Efficiency (η)	40.6% (Max)	81.2% (Max)	81.2% (Max)
Ripple Factor (γ)	$\sqrt{\{(V_{rms}/V_{dc})^2 - 1\}} = 1.21$	0.482	0.482
Peak Inverse Voltage (PIV)	V_m	$2 V_m$ (Requires expensive diodes)	V_m (Cost-effective)
Center-Tapped Transformer?	No	Yes (Bulky and expensive)	No (Operates on standard transformer)

Zener Diode Shunt Voltage Regulator: Heavily doped diode exhibiting sharp reverse breakdown at voltage V_Z . In parallel with load R_L , it absorbs line voltage variations and load current fluctuations: $I_S = I_Z + I_L = (V_{in} - V_Z) / R_S$, maintaining constant output $V_o = V_Z$.

2. Bipolar Junction Transistor (BJT) Characteristics & Operating Modes

A current-controlled three-terminal semiconductor device. Base current I_B modulates collector current $I_C = \beta I_B$. Current equation: $I_E = I_B + I_C$.

BJT Configuration	Input Current / Output Current	Input / Output Impedance	Voltage & Current Gain	Primary Application
Common Base (CB)	In: I_E Out: I_C	Very Low ($\sim 20 \Omega$) / Very High ($\sim 1 M\Omega$)	High Voltage Gain Current Gain $\alpha < 1$	High-frequency RF impedance matching
Common Emitter (CE)	In: I_B Out: I_C	Moderate ($\sim 1 k\Omega$) / Moderate ($\sim 50 k\Omega$)	High Voltage Gain High Current Gain β (20–500)	Audio & Voltage Amplifiers (Highest Power Gain)
Common Collector (CC)	In: I_B Out: I_E	Very High ($\sim 500 k\Omega$) / Very Low ($\sim 50 \Omega$)	Voltage Gain ≈ 1 High Current Gain γ	Emitter Follower / Buffer impedance bridging

UNIT IV: DIGITAL ELECTRONICS, BOOLEAN ALGEBRA & K-MAP MINIMIZATION

Number Bases, Binary Arithmetic, Universal Gates, De Morgan's Laws & 4-Variable K-Map Design

1. Number Systems, Binary Codes & 2's Complement Arithmetic

Bases: Decimal (Base 10: 0–9), Binary (Base 2: 0,1), Octal (Base 8: 0–7), Hexadecimal (Base 16: 0–9, A–F where A=10, B=11, C=12, D=13, E=14, F=15).

Binary Arithmetic & 2's Complement:

- 1's Complement: Invert every bit ($0 \rightarrow 1, 1 \rightarrow 0$).
- 2's Complement: $2's \text{ Complement} = 1's \text{ Complement} + 1$.
- Subtraction ($M - N$) using 2's Complement: Add 2's complement of N to M. If end carry occurs, discard carry (result is positive). If no carry occurs, take 2's complement of sum and attach negative sign.

Codes: BCD (Binary Coded Decimal: 8421 weights per digit), Excess-3 (BCD + 0011; self-complementing code), Gray Code (Unit-distance code where only one bit changes between successive numbers; prevents hardware glitches in shaft encoders).

2. Boolean Algebra Laws, Universal Gates & Karnaugh Maps (K-Maps)

De Morgan's Theorems:

Theorem 1: $(A + B)' = A' \cdot B'$ (The complement of a sum equals product of complements).

Theorem 2: $(A \cdot B)' = A' + B'$ (The complement of a product equals sum of complements).

Universal Logic Gates: NAND and NOR gates are universal because any Boolean function (AND, OR, NOT, XOR, XNOR) can be synthesized exclusively using either gate type alone.

4-Variable K-Map Minimization Rules:

1. Organize grid in Gray code order (00, 01, 11, 10) to maintain single-bit transitions between adjacent cells.
2. Identify cells with minterms (1s) and Don't Cares (X).
3. Form groups in descending powers of 2 (Octets of 8, Quads of 4, Pairs of 2). Larger groups eliminate more literals!
4. Wrap around edges: Corners (m_0, m_2, m_8, m_{10}) form a valid Quad.

MMDU END-SEMESTER EXAMINATION PAPER (SOLVED WITH DUAL DIAGRAMS)

Batch 2025–2029 • BELE-001: Basic of Electrical & Electronics Engineering • Total Marks: 60 • 100% Dual Diagrams

SECTION A: 12 COMPULSORY CONCEPTUAL QUESTIONS (1 MARK EACH = 12 MARKS)

Question 1 (Unit 1): Physical Meaning of Power Factor

1 Mark

Explain why low power factor is detrimental to industrial electrical transmission networks.

Question Power Triangle: Apparent Power S (kVA) = Active Power P (kW) + j Reactive Power Q (kVAR).

STEP-BY-STEP SOLUTION:

Solution Current Vector Magnification:

$I = P / (V \cos\phi)$. As $\cos\phi \downarrow \Rightarrow I \uparrow \Rightarrow I^2 R$ Losses $\uparrow$.

For a fixed real power P , drawing at low power factor forces current I to increase inversely. This magnifies $I^2 R$ transmission line heating losses, induces excessive line voltage drops, and demands larger conductor cross-sections and transformer ratings.

Question 2 (Unit 1): Resonance Condition in Series RLC Circuit

1 Mark

State why the net impedance of a series RLC circuit is purely resistive at resonant frequency.

Question Reactance Cancellation: Inductive Reactance $+jX_L$ balances Capacitive Reactance $-jX_C$.

STEP-BY-STEP SOLUTION:

Solution Phasor Cancellation Model:

$Z = R + j \frac{\omega L - 1}{\omega C}$. At $\omega_0 = 1/\sqrt{LC} \Rightarrow X_L = X_C \Rightarrow Z = R + j0 = R$.

At resonance, the counter-phase voltages across the inductor and capacitor cancel each other exactly ($V_L = -V_C$). The network behaves as a pure resistance with minimum impedance and unity power factor.

Question 3 (Unit 1): Maximum Power Transfer Efficiency Paradox

1 Mark

Why is the Maximum Power Transfer Theorem not applied to commercial electrical power utility grids?

Question Efficiency vs Transfer: Utility Goal = 95–98% Efficiency | MPT Goal = Maximum Watts to Load (50% Efficiency).

STEP-BY-STEP SOLUTION:

Solution Energy Loss Formulation:

Efficiency $\eta = P_L / (P_L + P_{int}) = R_L / (R_L + R_{th}) = 50\%$ when $R_L = R_{th}$.

Operating a utility grid at maximum power transfer wastes 50% of generated power as internal heat inside generators and transmission lines. Power systems operate at maximum efficiency ($R_L \gg R_{internal}$) rather than maximum power transfer.

Question 4 (Unit 2): Hysteresis vs. Eddy Current Loss Frequency Dependence

1 Mark

How do core losses change if a 50 Hz transformer is operated on a 60 Hz supply at constant voltage?

Question Frequency Variation: V/f ratio changes flux density $B_m \propto V/f$.

STEP-BY-STEP SOLUTION:

Solution Loss Scaling Proportions:

$P_h \propto f \cdot (V/f)^{1.6} = V^{1.6} f^{-0.6} \Rightarrow \text{ext}\{\text{Decreases!}\}$

$P_e \propto f^2 \cdot (V/f)^2 = V^2 \Rightarrow \text{ext}\{\text{Remains Constant!}\}$

At constant voltage, flux density decreases as frequency rises. Hysteresis loss decreases slightly (proportional to $f^{-0.6}$) while eddy current loss remains constant, causing total core losses to decrease.

Question 5 (Unit 2): Significance of Back EMF in DC Motors

1 Mark

Why does a DC motor draw dangerously high current if started without an external starter?

Question Starting Circuit: At $N = 0 \Rightarrow$ Back EMF $E_b = 0 \Rightarrow$ Armature Current $I_a = V / R_a$.

STEP-BY-STEP SOLUTION:

Solution Current Surge Formula:

$$I_{a, start} = (V - 0) / R_a = 230 \text{ V} / 0.5 \text{ } \Omega = 460 \text{ A} \approx 15\text{--}20 \text{ times } I_{rated}!$$

At standstill ($N=0$), back EMF is zero. Because armature resistance R_a is extremely small ($\sim 0.5 \text{ } \Omega$), direct connection to rated voltage produces a destructive current surge that will burn the windings unless limited by a starter.

Question 6 (Unit 2): Slip Condition of Induction Motor at Synchronous Speed

1 Mark

Why can an induction motor never run at synchronous speed ($N = N_s$)?

Question Relative Velocity: Rotor speed N vs Stator RMF speed $N_s \Rightarrow$ Relative motion $\Delta N = N_s - N$.

STEP-BY-STEP SOLUTION:

Solution Induced EMF Zero Vector:

If $N = N_s \Rightarrow$ Relative Speed $= 0 \Rightarrow \text{ext}\{d\Phi/dt\} = 0 \Rightarrow E_r = 0 \Rightarrow I_r = 0 \Rightarrow T = 0!$

If the rotor reached synchronous speed, rotor conductors would not cut any rotating magnetic flux. Induced rotor EMF, current, and electromagnetic torque would drop to zero, causing the rotor to immediately decelerate.

Question 7 (Unit 3): Barrier Potential of Silicon vs. Germanium

1 Mark

Why is the forward knee voltage of Silicon (0.7V) higher than that of Germanium (0.3V)?

Question Bandgap Atomic Dimension: Si ($E_g = 1.1 \text{ eV}$, tightly bound valence electrons) vs Ge ($E_g = 0.67 \text{ eV}$).

STEP-BY-STEP SOLUTION:

Solution Barrier Height Equation:

$$V_o = V_T \ln(N_A N_D / n_i^2).$$
 Higher bandgap in Si yields lower intrinsic carrier density $n_i \Rightarrow$ higher built-in potential V_o .

Silicon's larger energy bandgap (1.1 eV vs 0.67 eV) creates a stronger internal electric field across the depletion region, requiring 0.7V of forward bias to overcome compared to 0.3V for Germanium.

Question 8 (Unit 3): Peak Inverse Voltage in Bridge vs. Center-Tapped Rectifier

1 Mark

Compare the Peak Inverse Voltage (PIV) rating demanded of diodes in Bridge vs Center-Tapped rectifiers.

Question Reverse Voltage Stress: Bridge Diode sees V_m | Center-Tapped Diode sees $2 V_m$.

STEP-BY-STEP SOLUTION:

Solution PIV Ratio Formulation:

$$PIV_{bridge} = V_m \text{ versus } PIV_{center-tap} = 2 V_m.$$

In a center-tapped full-wave rectifier, each non-conducting diode must withstand the full secondary voltage $2 V_m$. In a bridge rectifier, the reverse voltage divides across two reverse diodes, requiring only V_m rating.

Question 9 (Unit 3): BJT Base Width Design Principle

1 Mark

Why is the base region of a BJT manufactured ultra-thin and lightly doped?

Question Carrier Diffusion Path: Emitter (Heavy Injection) $\rightarrow$ Base (Minimal Recombination) $\rightarrow$ Collector (Extraction).

STEP-BY-STEP SOLUTION:

Solution Recombination Minimization Model:

Base Recombination $< 2\% \Rightarrow IC \approx IE \Rightarrow \alpha \approx 1 \Rightarrow \alpha = IC/IE \approx 0.98\text{--}0.99.$

A thin, lightly doped base ensures that almost all injected minority carriers traverse the base without recombining, allowing over 98% of carriers to be swept into the collector for high current gain β .

Question 10 (Unit 4): Self-Complementing Property of Excess-3 Code**1 Mark**

Demonstrate why Excess-3 code is termed a self-complementing code using decimal digit 4.

Question 9's Complement Inversion: 9's complement of decimal 4 is $(9 - 4) = 5$.

STEP-BY-STEP SOLUTION:**Solution Bitwise Inversion Trace:**

$\text{Excess-3}(4) = 4 + 3 = 7 = 0111_2$.

$\text{Bitwise NOT}(0111_2) = 1000_2 = 8_{10} = 5 + 3 = \text{Excess-3}(5)$!

Taking the 1's complement of an Excess-3 codeword directly produces the Excess-3 representation of its 9's complement, greatly simplifying digital decimal subtraction arithmetic.

Question 11 (Unit 4): Why NAND is Termed a Universal Gate**1 Mark**

Prove that a NOT gate can be synthesized using a single 2-input NAND gate.

Question Input Tie-Together: Connect both inputs of NAND together to input signal A.

STEP-BY-STEP SOLUTION:**Solution Boolean Idempotence Formulation:**

$Y = (A \cdot A)' = A'$ (By Boolean Idempotent Law).

By connecting both inputs of a 2-input NAND gate to common signal A, the output becomes $(A \cdot A)' = A'$, functioning identically to an inverter (NOT gate).

Question 12 (Unit 4): Gray Code Usage in Rotary Optical Encoders**1 Mark**

Why is Gray code preferred over standard Binary code in optical shaft position encoders?

Question Binary Transition Glitch: Binary 7 (0111) $\rightarrow$ 8 (1000) flips 4 bits simultaneously; Gray flips only 1 bit.

STEP-BY-STEP SOLUTION:**Solution Unit Distance Property:**

Hamming Distance = 1 between adjacent counts $\Rightarrow$ eliminates intermediate spurious read codes.

In standard binary, multiple bits switch simultaneously (e.g. 0111 to 1000). Due to physical sensor misalignment, bits switch at slightly different nanoseconds, creating erroneous intermediate spikes. Gray code changes only one bit at a time, preventing read errors.

SECTION B: 4 ANALYTICAL QUESTIONS (2 MARKS EACH = 8 MARKS)

Question 13 (Unit 1): Thevenin Equivalent Circuit Derivation

2 Marks

Find the Thevenin equivalent circuit across terminals A-B for a circuit with a 24V DC source in series with 6Ω, connected across a parallel 12Ω resistor.

Question Voltage Divider Topology: 24V source → 6Ω resistor → Node A [12Ω connected to Node B/ground].

STEP-BY-STEP SOLUTION:

Solution Thevenin Derivation Equations:

$$V_{th} = V_{oc} = 24 \times \left[\frac{12}{6 + 12} \right] = 24 \times \frac{12}{18} = 16 \text{ V}$$
$$R_{th} = 6 \parallel 12 = \frac{6 \times 12}{6 + 12} = 72 / 18 = 4 \text{ } \Omega$$

The entire circuit across terminals A-B simplifies to a Thevenin voltage source $V_{th} = 16 \text{ V}$ in series with a resistance $R_{th} = 4 \text{ } \Omega$.

Question 14 (Unit 2): Open-Circuit and Short-Circuit Test Diagnostic Roles

2 Marks

Explain why the wattmeter reading in an Open-Circuit (OC) test represents core loss only, while in a Short-Circuit (SC) test it represents copper loss only.

Question Diagnostic Regimes: OC Test [Rated Voltage, Negligible No-Load Current I_0] vs SC Test [Rated Current, Tiny Voltage V_{sc}].

STEP-BY-STEP SOLUTION:

Solution Decoupled Loss Models:

OC Test: $I_0 \approx 2\text{--}5\% \times I_{rated} \Rightarrow I_0^2 R_{Cu} \approx 0 \Rightarrow W_0 = P_{iron}$

SC Test: $V_{sc} \approx 5\text{--}10\% \times V_{rated} \Rightarrow B_m \approx 0 \Rightarrow P_{iron} \approx 0 \Rightarrow W_{sc} = P_{Cu}$

In the OC test, current is so low that copper loss is negligible. In the SC test, the applied voltage is tiny, so core flux and core loss are negligible, isolating the respective loss mechanisms perfectly.

Question 15 (Unit 3): Zener Shunt Regulator Load Current Calculation

2 Marks

A 10V Zener regulator has input $V_{in} = 20V$, series resistor $R_S = 200\Omega$, and load $R_L = 1 \text{ k}\Omega$. Calculate source current I_S , load current I_L , and Zener current I_Z .

Question Shunt Circuit Diagram: 20V DC → R_S (200Ω) → Node V_Z (10V) → Parallel branches: Zener Diode & R_L (1kΩ).

STEP-BY-STEP SOLUTION:

Solution Current Distribution Formulation:

$$I_S = (V_{in} - V_Z) / R_S = (20 - 10) / 200 = 10 / 200 = 0.05 \text{ A} = 50 \text{ mA}$$
$$I_L = V_Z / R_L = 10 \text{ V} / 1000 \text{ } \Omega = 0.01 \text{ A} = 10 \text{ mA}$$
$$I_Z = I_S - I_L = 50 \text{ mA} - 10 \text{ mA} = 40 \text{ mA}$$

The Zener diode safely conducts 40 mA, maintaining a steady 10V regulated output across the 1 kΩ load.

Question 16 (Unit 4): 2's Complement Subtraction

2 Marks

Perform subtraction $(28)_{10} - (15)_{10}$ using 8-bit 2's complement arithmetic.

Question Binary Operand Conversion: $28 = 00011100_2 \mid 15 = 00001111_2$.

STEP-BY-STEP SOLUTION:

Solution 2's Complement Addition Trace:

1's Complement of 15: 11110000_2

2's Complement of 15: $11110000 + 1 = 11110001_2$

Add to 28: $00011100 + 11110001 = 1\ 00001101_2$

Discard End Carry (1) ⇒ Result = $00001101_2 = 8 + 4 + 1 = (13)_{10}$ (Correct!)

The end carry indicates a positive result, yielding $00001101_2 = 13_{10}$, matching $28 - 15 = 13$.

SECTION C: 4 COMPREHENSIVE PROBLEMS (4 MARKS EACH = 16 MARKS)

Question 17 (Unit 1): Complete AC Series RLC Circuit Analysis

4 Marks

A coil of resistance $R = 10\Omega$ and inductance $L = 0.1H$ is connected in series with capacitor $C = 50\mu F$ across a 230V, 50Hz AC supply. Calculate: (1) Inductive & capacitive reactance, (2) Total impedance, (3) Current and power factor, (4) Active and reactive power.

Question Series RLC Schematic: 230V, 50Hz $\rightarrow$ R (10Ω) $\rightarrow$ L ($0.1H$) $\rightarrow$ C ($50\mu F$).

STEP-BY-STEP SOLUTION:

Solution Step-by-Step Impedance & Power Calculations:

- Reactances: $X_L = 2\pi f L = 2 \times 3.1416 \times 50 \times 0.1 = 31.42 \text{ } \Omega$
 $X_C = 1 / (2\pi f C) = 1 / (2 \times 3.1416 \times 50 \times 50 \times 10^{-6}) = 63.66 \text{ } \Omega$
 Net Reactance $X = X_L - X_C = 31.42 - 63.66 = -32.24 \text{ } \Omega$ (Capacitive!)
- Impedance: $|Z| = \sqrt{R^2 + X^2} = \sqrt{10^2 + (-32.24)^2} = \sqrt{100 + 1039.4} = 33.76 \text{ } \Omega$
- Current: $I = V / |Z| = 230 / 33.76 = 6.813 \text{ A}$
- Power Factor: $\cos\phi = R / |Z| = 10 / 33.76 = 0.2962$ (Leading)
- Active Power: $P = V I \cos\phi = 230 \times 6.813 \times 0.2962 = 464.15 \text{ W}$
- Reactive Power: $Q = V I \sin\phi = 230 \times 6.813 \times \sqrt{1 - 0.2962^2} = 1494.5 \text{ VAR (Leading)}$

Because capacitive reactance dominates ($X_C > X_L$), the circuit draws a leading current of 6.81 A at a power factor of 0.296 leading.

Question 18 (Unit 2): Transformer Efficiency and All-Day Efficiency

4 Marks

A 50 kVA distribution transformer has iron loss $P_i = 500W$ and full-load copper loss $P_{cu} = 800W$. Calculate: (1) Full-load efficiency at 0.8 pf lagging, (2) Load kVA at which maximum efficiency occurs.

Question Loss Energy Flow: Input Power $\rightarrow$ [Iron Loss 500W + Copper Loss 800W] $\rightarrow$ Output Power = kVA $\times$ pf.

STEP-BY-STEP SOLUTION:

Solution Efficiency Formulations:

- Full-Load Output = 50 kVA $\times$ 0.8 = 40 kW = 40,000 W
 Total Full-Load Losses = $P_i + P_{cu} = 500 + 800 = 1300 \text{ W}$
 Efficiency $\eta_{(fl)} = [40000 / (40000 + 1300)] \times 100 = \frac{40000}{41300} \times 100 = 96.85\%$
- Load fraction for Maximum Efficiency: $x = \sqrt{P_i / P_{cu, f_1}} = \sqrt{\frac{500}{800}} = \sqrt{0.625} = 0.7906$
 Load kVA for Max Efficiency = $x \times 50 \text{ kVA} = 0.7906 \times 50 = 39.53 \text{ kVA}$

Maximum efficiency occurs when the transformer is loaded to **39.53 kVA** (79% of rated capacity), where variable copper loss equals 500W.

Question 19 (Unit 3): Derivation of Full-Wave Bridge Rectifier Ripple Factor

4 Marks

Derive the mathematical expression for Ripple Factor γ of a Full-Wave Bridge Rectifier and prove it equals 0.482.

Question Rectified Waveform: Pulsating DC with period $T/2$: $v(t) = |V_m \sin(\omega t)|$.

STEP-BY-STEP SOLUTION:

Solution Ripple Factor Derivation Trace:

$$\begin{aligned} \text{Ripple Factor } \gamma &= V_{aC, rms} / V_{dc} = \sqrt{(V_{rms}/V_{dc})^2 - 1} \\ V_{dc} &= \frac{1}{\pi} \int_0^\pi V_m \sin(\theta) d\theta = (2 V_m) / \pi \approx 0.6366 V_m \\ V_{rms} &= \sqrt{\frac{1}{\pi} \int_0^\pi V_m^2 \sin^2(\theta) d\theta} = V_m / \sqrt{2} \approx 0.7071 V_m \\ \text{Form Factor } K_f &= V_{rms} / V_{dc} = \frac{V_m / \sqrt{2}}{2 V_m / \pi} = \pi / (2 \sqrt{2}) \approx 1.1107 \\ \gamma &= \sqrt{(1.1107)^2 - 1} = \sqrt{1.2337 - 1} = \sqrt{0.2337} = 0.4834 \approx 0.482 \end{aligned}$$

The ripple factor evaluates to 0.482, indicating that AC harmonic content is less than half of the DC component without filtering.

Minimize the Boolean function: $F(A, B, C, D) = \Sigma m(1, 3, 7, 11, 15) + d(0, 2, 5)$ using a 4-variable K-Map. Implement using basic gates.

Question K-Map Cell Mapping: Minterms at 1, 3, 7, 11, 15; Don't cares at 0, 2, 5.

STEP-BY-STEP SOLUTION:

Solution Optimal Grouping Strategy:

Group 1 (Octet): Combine minterms $m_1, m_3, m_7, m_{11}, m_{15}$ with don't cares d_0, d_2, d_5 !

Cells (0, 1, 2, 3, 5, 7, 11, 15): Notice column CD = (m_3, m_7, m_{11}, m_{15}) forms a Quad CD

Row A'B' (0, 1, 2, 3) + (5, 7): Combine (1, 3, 5, 7) with (3, 7, 11, 15) $\Rightarrow$

Quad 1: (3, 7, 11, 15) yields CD

Quad 2: (1, 3, 5, 7) yields A'D

Quad 3: (0, 1, 2, 3) with d_0, d_2 yields A'B'

Minimal SOP Expression: $F = CD + A'B' + A'D$ (or $F = CD + A'B'$)

By leveraging Don't Care cells $d(0, 2)$, the function simplifies down to $F = CD + A'B'$, eliminating 3 gates.

SECTION D: 4 EXTENDED ESSAY & DESIGN PROBLEMS (6 MARKS EACH = 24 MARKS)

Question 21 (Unit 1): Superposition and Thevenin Verification on Multi-Source Network

6 Marks

For a network with two DC voltage sources $V_1 = 30\text{V}$ (with internal resistance 2Ω) and $V_2 = 20\text{V}$ (with internal resistance 4Ω) connected in parallel across a common load resistor $R_L = 8\Omega$. Find current through R_L using: (1) Superposition Theorem, (2) Thevenin's Theorem. Verify both methods give identical results.

Question Parallel Source Bridge Topology: Branch 1 (30V , 2Ω) $\parallel$ Branch 2 (20V , 4Ω) $\rightarrow$ Load R_L (8Ω).

STEP-BY-STEP SOLUTION:

Solution Step-by-Step Dual Theorem Verification:

Method 1: Superposition Theorem

Case A: V_1 (30V) active, V_2 shorted:

$$R_{eq,1} = 2 + (4 \parallel 8) = 2 + \frac{32}{12} = 2 + 2.667 = 4.667 \text{ } \Omega \Rightarrow I_1 = 30 / 4.667 = 6.429 \text{ A}$$

$$\text{Current through } R_L: IL_1 = 6.429 \text{ times } [4 / (4 + 8)] = 6.429 \text{ times } \frac{4}{12} = 2.143 \text{ A } (\downarrow)$$

Case B: V_2 (20V) active, V_1 shorted:

$$R_{eq,2} = 4 + (2 \parallel 8) = 4 + \frac{16}{10} = 5.60 \text{ } \Omega \Rightarrow I_2 = 20 / 5.60 = 3.571 \text{ A}$$

$$\text{Current through } R_L: IL_2 = 3.571 \text{ times } [2 / (2 + 8)] = 3.571 \text{ times } \frac{2}{10} = 0.714 \text{ A } (\downarrow)$$

$$\text{Total Load Current: } IL = IL_1 + IL_2 = 2.143 + 0.714 = 2.857 \text{ A}$$

Method 2: Thevenin's Theorem

$$\text{Open circuit terminals of } R_L: V_{th} = V_2 + (V_1 - V_2) \text{ times } [4 / (2 + 4)] = 20 + 10 \text{ times } \frac{4}{6} = 26.67 \text{ V}$$

$$R_{th} = 2 \parallel 4 = \frac{2 \text{ times } 4}{2 + 4} = 8 / 6 = 1.333 \text{ } \Omega$$

$$\text{Load Current: } IL = V_{th} / (R_{th} + R_L) = 26.67 / (1.333 + 8) = 26.67 / 9.333 = 2.857 \text{ A}$$

Both Superposition and Thevenin theorems converge to an identical load current of **2.857 A**.

Question 22 (Unit 2): Complete Theory and Speed Control of DC Shunt Motor

6 Marks

Derive the Torque equation of a DC Motor. Describe the two methods of speed control for a DC Shunt Motor (Armature Resistance Control and Field Flux Control) with relevant circuit diagrams, equations, and speed-torque characteristics.

Question Speed-Torque Characteristic Slopes: Rated speed $N_{rated} \rightarrow$ Armature control drops speed below N_{rated} ; Field control raises speed above N_{rated} .

STEP-BY-STEP SOLUTION:

Solution Torque & Speed Derivations:

1. Electrical Power converted to Mechanical: $P_m = E_b I_a = T \cdot \omega$

$$\text{Substitute } E_b = \frac{\Phi Z N P}{60 A} \text{ and } \omega = 2\pi N / 60:$$

$$T \cdot \frac{2\pi N}{60} = \left[\frac{\Phi Z N P}{60 A} \right] \cdot I_a \Rightarrow T = \left[\frac{P Z}{2\pi A} \right] \cdot \Phi I_a = 0.159 \Phi Z I_a \text{ (P/A) N}\cdot\text{m}$$

$$2. \text{ Speed Governing Relation: } N = \frac{V - I_a R_a}{k \Phi}$$

Method A (Armature Resistance): R_{ex} added in series with $R_a \Rightarrow N \downarrow$ below rated speed. Wasteful $I^2 R$ loss.

Method B (Field Flux): R_f added in series with shunt field $\Rightarrow I_f \downarrow \Rightarrow \Phi \downarrow \Rightarrow N \uparrow$ above rated speed. Highly efficient!

Field control provides a constant-horsepower drive above base speed, whereas armature control provides a constant-torque drive below base speed.

Question 23 (Unit 3): BJT Common Emitter Input and Output Characteristics

6 Marks

Sketch and thoroughly explain the Input and Output static characteristics of an NPN Bipolar Junction Transistor in Common Emitter (CE) configuration. Define Early Effect (Base-Width Modulation) and explain how the BJT acts as a digital switch.

Question Characteristic Plots: Input (I_B vs V_{BE} at constant V_{CE}) | Output (I_C vs V_{CE} at constant I_B showing Cutoff, Active, and Saturation).

STEP-BY-STEP SOLUTION:

Solution 3 Operating Regimes & Early Effect:

1. Cutoff ($V_{BE} \leq 0$, $V_{CE} = V_{CC}$): Both junctions reverse biased; $I_C \approx 0$ $\Rightarrow$ Switch OFF (Open)

2. Saturation ($V_{BE} > 0.7\text{V}$, $V_{CE} \approx 0.2\text{V}$): Both junctions forward biased; $I_C = I_{C,sat}$ $\Rightarrow$ Switch ON (Closed)

3. Active ($V_{BE} > 0.7\text{V}$, $V_{CE} > 1\text{V}$): EBJ forward, CBJ reverse; linear relation $I_C = \beta I_B \Rightarrow$ Linear Amplifier

Early Effect: As $V_{CE} \uparrow \Rightarrow$ Reverse bias on CBJ $\uparrow \Rightarrow$ Depletion width $\uparrow \Rightarrow$ Effective base width $W_b \downarrow \Rightarrow I_C$ slopes upward

The Early Effect causes collector current curves to have a slight positive slope in the active region; projecting these curves backward intercepts the negative voltage axis at the Early Voltage $-V_A$.

Design and implement a 1-bit Full Adder using only 2-input NAND gates. Derive the Sum and Carry equations, provide the complete truth table, and draw the universal NAND gate schematic diagram.

Question Adder Signal Interface: Inputs A, B, C_{in} → Combinational Logic → Outputs Sum S and Carry C_{out}.

STEP-BY-STEP SOLUTION:

Solution Full Adder NAND Synthesis Trace:

1. Boolean Equations: Sum $S = A \oplus B \oplus C_{in}$ | Carry $C_{out} = AB + BC_{in} + AC_{in}$

2. Half Adder with NAND: A 2-input XOR requires exactly 4 NAND gates:
 $N1 = (AB)'$ | $N2 = (A \cdot N1)'$ | $N3 = (B \cdot N1)'$ | $N4 = (N2 \cdot N3)' = A \oplus B$
Carry generation: $N5 = (N1)' = AB$

3. Cascaded Full Adder Architecture: Two cascaded Half Adders + 1 final OR gate:
Total NAND gates required = 4 (HA1) + 4 (HA2) + 1 (Combine carries) = 9 NAND Gates Total!

This 9-NAND gate architecture produces both the Sum and Carry outputs using a single integrated circuit type, optimizing silicon die manufacturing.